Data Visualization Using Tableau

Question Bank with Answers

Part A - 2 Marks

Q1. Define a Tableau dashboard. State why dashboards are used in organisations.

Introduction: A Tableau dashboard is an interactive workspace that combines several worksheets, filters, text, images and other objects into one consolidated view, so that related information can be monitored together.

- **Definition and components:** A dashboard collects worksheets such as Total Sales, Region-wise Sales and Monthly Trend on a single screen along with KPI cards, legends, filters and text objects. Users interact with one view instead of opening each worksheet separately.
- **Why dashboards are used:** Organisations generate data from many sources and separate reports are slow to read. A dashboard centralises KPIs, supports comparison and trend identification, reduces reporting time and helps managers detect problems such as falling sales or low attendance early.

Q2. What is a Tableau Story? Explain story points, titles and captions.

Introduction: A Tableau Story is a sequence of worksheets and dashboards arranged as story points to communicate analytical findings in a logical order, from the overall situation to the conclusion.

- **Story and story points:** The story is the complete presentation and each story point is one stage that holds a worksheet or dashboard, for example overall sales, monthly trend, regional sales, profit and conclusion. New points are added using New Blank Point and arranged in order.
- **Titles and captions:** A title names the subject of a story point, whereas a caption explains the insight, such as "North region contributed the largest share of revenue". A story is saved inside the workbook (.twb or .twbx), not as a separate file.

Q3. Explain ad hoc reporting in Tableau. Write the calculated field for Profit Ratio and state its output when Sales is ₹80,000 and Profit is ₹12,000.

Introduction: Ad hoc reporting means creating a report on demand to answer one particular question, by choosing dimensions, measures, filters and chart types freely instead of using a fixed predefined format.

- **Ad hoc report:** It is on-demand, flexible and question-oriented; for example, a manager asks which product earned the highest profit in the West region and the analyst builds the view immediately. It differs from a standard report, which is predefined and repeated every period.
- **Profit Ratio calculated field:** The calculation below aggregates Profit and Sales first and then divides, so the ratio remains correct at any level of detail. Expected output: (12,000 / 80,000) x 100 = 15%. Do not multiply by 100 again if Tableau's percentage number format is applied.

```
SUM([Profit]) / SUM([Sales]) * 100
```

Q4. Define drill-down. Explain how a Geographic Hierarchy is created and used in Tableau.

Introduction: Drill-down is an analysis technique that moves from summarised values to progressively lower levels of detail through a hierarchy of related dimensions.

- **Meaning and example:** A user starts with Country-level sales, expands to State-level sales and then to City-level sales, for instance India to Karnataka to Bengaluru. Child-level values must reconcile with their parent totals under the same filters and aggregation.
- **Creating and using the hierarchy:** In the Data pane right-click Country, choose Hierarchy > Create Hierarchy, name it and drag State and City below it. Place the hierarchy on Rows with Sales, then use the plus and minus controls to expand or collapse levels.

Q5. What is drill-through reporting? Name the actions required to implement it in Tableau.

Introduction: Drill-through is a reporting technique in which the user selects a summarised value and opens a separate detail view containing the records that belong to that selection.

- **Meaning and example:** A regional summary shows total sales of each region; selecting West opens a transaction report listing Order ID, customer, product, sales and profit only for West. It is useful for audit, investigation of unusual values and review of individual records.
- **Implementation:** Use Dashboard > Actions > Add Action > Filter to pass the selected Region to the detail sheet, and a Go to Sheet action to navigate to the detail dashboard. Go to Sheet alone does not filter the data, so both actions are configured and a return path is provided.

Q6. A shop records monthly sales (₹ thousands) from January to June as 18, 24, 22, 30, 36 and 45. Calculate the percentage increase from January to June and state the steps to draw the line graph in Tableau.

Introduction: A line graph connects data points in time order and is used to show the trend of a measure over a period.

- **Percentage increase:** Percentage increase = [(Final - Initial) / Initial] x 100 = [(45 - 18) / 18] x 100 = (27 / 18) x 100 = 150%. Sales therefore increased by 150% from January to June, with a small dip in March (24 to 22) before rising steadily.
- **Steps in Tableau:** Connect to the data source, drag Month to Columns and Sales to Rows, and select Line on the Marks card or Show Me > Line Chart. Keep months in calendar order, add Sales to Label to show values and format the axis as currency in thousands.

Q7. A company records sales (₹ thousands) of Products A, B and C as follows. Calculate the total sales of each quarter and explain how a stacked bar chart is created.

Quarter	Product A	Product B	Product C
Q1	15	25	20
Q2	20	30	25
Q3	35	20	30

Introduction: A stacked bar chart divides each bar into segments, showing both the total of a category and the contribution of each sub-category to that total.

- **Quarter totals:** Q1 = 15 + 25 + 20 = 60; Q2 = 20 + 30 + 25 = 75; Q3 = 35 + 20 + 30 = 85 (₹ thousands). Total sales rise every quarter, and Q3 is the highest with Product A contributing the largest segment (35).
- **Steps in Tableau:** Pivot the three product columns into two fields, Product and Sales. Drag Quarter to Columns and Sales to Rows, choose Bar (or Show Me > Stacked Bars), drag Product to Color and ensure Analysis > Stack Marks > On is selected.

Q8. An online store records 8,000 website visitors, 5,000 product views, 2,400 add-to-cart events and 1,200 completed purchases. Calculate the overall conversion and drop-off rate and identify the stage with the largest loss.

Introduction: A funnel chart shows the number of records remaining at each successive stage of a process, making drop-offs between stages easy to identify.

- **Overall conversion and drop-off:** Conversion rate = (1,200 / 8,000) x 100 = 15%. Drop-off rate = 100% - 15% = 85%, meaning 85% of the visitors did not complete a purchase. Stage-wise conversion is 62.5% (views), 48% (cart) and 50% (purchase).
- **Largest loss and construction:** Drop-off is 37.5% from visitors to views, 52% from views to cart and 50% from cart to purchase, so the largest loss is between product view and add-to-cart. In Tableau, place Stage on Rows, count on Columns, sort by process order and use bars.

Q9. Define a histogram. How does it differ from a bar chart, and how is it created in Tableau?

Introduction: A histogram displays the distribution of a continuous numerical variable by grouping observations into intervals called bins, where bar height shows the number of observations in each bin.

- **Difference from a bar chart:** A bar chart compares separate categories such as regions, whereas a histogram represents continuous numerical intervals such as marks 0-10, 10-20. It reveals the centre, spread, skewness and gaps of the data, and its shape depends on the chosen bin width.
- **Creation in Tableau:** Right-click the numeric field (for example Marks) in the Data pane and choose Create > Bins with a size such as 10. Drag the bin field to Columns, drag Number of Records to Rows and select Bar marks; the Y axis should count records, not sum the field.

Q10. What is a Gantt chart? Write the calculated field used to find task duration and explain how the chart is built.

Introduction: A Gantt chart is a time-based visualisation in which each task is a horizontal bar whose starting position shows the start date and whose length shows the task duration.

- **Duration field:** When the data has start and end dates but no duration, the calculation below returns the number of days between them and is used to control bar length. Expected output: a start date of 01-Mar and an end date of 11-Mar gives a duration of 10 days.
- **Construction:** Drag Start Date to Columns and Task Name to Rows, select Gantt Bar on the Marks card, drag Duration to Size and Status or Project to Color. Overlapping bars show parallel tasks, and gaps indicate waiting periods.

```
DATEDIFF('day', [Start Date], [End Date])
```

Part B - 5 Marks

Q11. Differentiate between a Dashboard and a Story in Tableau with an example.

Introduction: A dashboard is a consolidated, interactive view of several worksheets used for monitoring, while a story is a sequence of story points used to explain findings step by step.

Basis	Dashboard	Story
Meaning	Collection of multiple visualisations in one view	Sequence of visualisations or dashboards arranged as story points
Purpose	Interactive analysis and monitoring	Communicating findings step by step
Structure	Multiple worksheets in a single consolidated interface	Multiple story points, each holding a sheet or dashboard
Navigation	Filters and dashboard actions	Story-point navigation (tabs or numbers)
Typical use	Monitoring sales, profit and KPIs together	Explaining how and why sales changed over a period

Example: A Sales Performance Dashboard shows total sales, region-wise sales and category profit together for monitoring, whereas a Story on the same data presents overall sales, regional comparison, profit analysis and recommendations in sequence to a management audience.

Q12. Differentiate between Drill-down and Drill-through reporting with an example from the retail domain.

Introduction: Drill-down moves through the levels of a hierarchy inside the same view, whereas drill-through moves from a selected summary value to a separate detailed view or report.

Basis	Drill-down	Drill-through
Meaning	Moves from summary to lower hierarchical levels	Moves from a summary to a separate detail view or report
Structure	Hierarchy within one view	Source summary view and target detail view
Navigation	Expand or collapse levels using plus and minus	Select a value and navigate or filter
Typical detail	Aggregated values at lower levels	Individual records or transactions
Tableau implementation	Dimension hierarchies and date levels	Filter actions and Go to Sheet actions

Example: Drill-down: Category to Sub-Category to Product, where each expansion shows lower-level sales totals. Drill-through: selecting the Furniture category summary opens the list of individual customer orders for Furniture. Drill-down asks at which lower level the total is distributed; drill-through asks which records make up the selected total.

Q13. Differentiate between strategic, tactical, operational and analytical dashboards.

Introduction: Dashboards are classified by purpose, intended users and the kind of information displayed, and each type serves a different level of an organisation.

Basis	Strategic	Tactical	Operational	Analytical
Main purpose	Long-term planning	Departmental monitoring	Daily monitoring	Detailed analysis
Users	Executives	Department managers	Operational staff	Analysts
Information	Highly summarised KPIs	Department-specific	Current operational data	Detailed and historical
Time period	Long-term	Medium-term	Daily or near real-time	Historical and current
Example	University pass and placement percentage	Marketing campaign performance	Hospital beds and pending reports	Sales trend and profitability analysis

Example: A hospital management board uses a strategic dashboard for yearly outcomes, while the ward staff use an operational dashboard showing admitted patients and available beds each day.

Q14. A retailer records the following category-wise sales and profit. Calculate each category's contribution to total sales and its profit margin, identify the best category, and explain how the treemap is built in Tableau.

Category	Sales (₹)	Profit (₹)
Mobiles	90,000	13,500
Laptops	60,000	6,000
Wearables	30,000	7,500
Accessories	20,000	1,000
Total	2,00,000	28,000

Introduction: A treemap represents hierarchical data as nested rectangles whose area is proportional to one measure and whose colour represents a second measure.

- **Visual encoding:** Each rectangle is a category; its size shows Sales and its colour shows Profit. Labels display names and values and tooltips give details. This allows two measures to be compared in a compact space.
- **Sales contribution:** Contribution = (Category Sales / Total Sales) x 100. Mobiles = 90,000 / 2,00,000 = 45%; Laptops = 30%; Wearables = 15%; Accessories = 10%. Mobiles therefore occupies the largest rectangle, nearly half of the treemap area.
- **Profit margin:** Margin = (Profit / Sales) x 100. Mobiles = 13,500 / 90,000 = 15%; Laptops = 6,000 / 60,000 = 10%; Wearables = 7,500 / 30,000 = 25%; Accessories = 1,000 / 20,000 = 5%. The overall margin is 28,000 / 2,00,000 = 14%.
- **Tableau implementation:** Drag Category to Detail, Sales to Size, choose Treemap in Show Me, drag Profit to Color and add Category and Sales to Label. For margin, create SUM([Profit]) / SUM([Sales]), format it as a percentage and drag it to Color.
- **Interpretation and limitation:** Wearables is the best category because it has the highest margin (25%) despite only 15% of sales, while Accessories is weakest at 5%. Limitation: rectangles of similar size are hard to compare exactly and negative values are not shown by area.

Q15. The advertising expenditure and sales of a company for five months are given below. Calculate the Pearson correlation coefficient and explain how the relationship is shown using a scatter plot in Tableau.

Month	Advertising (X, ₹ thousand)	Sales (Y, ₹ lakh)
January	2	12
February	4	17
March	6	19
April	8	26
May	10	31

Introduction: A scatter plot displays the relationship between two numerical variables, where each mark is one observation positioned by its X and Y values; Pearson's r measures the strength and direction of the linear relationship.

- **Formula:** $r = [n(\Sigma XY) - (\Sigma X)(\Sigma Y)] / \sqrt{\{[n(\Sigma X^2) - (\Sigma X)^2]\ [n(\Sigma Y^2) - (\Sigma Y)^2]\}}$. The value of r lies between -1 and +1, where +1 is a perfect positive and -1 a perfect negative linear relationship.
- **Required sums:** With n = 5: $\Sigma X = 2+4+6+8+10 = 30$; $\Sigma Y = 12+17+19+26+31 = 105$; $\Sigma X^2 = 4+16+36+64+100 = 220$; $\Sigma Y^2 = 144+289+361+676+961 = 2431$; $\Sigma XY = 24+68+114+208+310 = 724$.
- **Substitution:** Numerator = 5(724) - 30(105) = 3620 - 3150 = 470. Denominator = $\sqrt{[(5 \times 220 - 900)(5 \times 2431 - 11025)]} = \sqrt{(200 \times 1130)} = \sqrt{226000} \approx 475.39$. Therefore $r = 470 / 475.39 \approx +0.989$.
- **Tableau steps:** Drag Advertising to Columns, Sales to Rows and Month to Detail, select Circle as mark type, and drag a Linear Trend Line from the Analytics pane. Color or Size may be added to represent extra measures.
- **Interpretation:** $r \approx +0.989$ shows a very strong positive linear relationship: higher advertising spend is associated with higher sales and the points rise from left to right. Correlation does not prove causation, and outliers can distort it.

Q16. The marks of three students in three subjects are given below. Explain how a heat map is created in Tableau, calculate the average marks and identify the highest and lowest marks.

Student	Data Structures	DBMS	Networks
P	78	85	64
Q	90	70	82
R	66	74	88

Introduction: A heat map uses colour intensity to represent the magnitude of a numerical measure across the combinations of two categorical dimensions, so high and low values are seen at a glance.

- **Heat map structure:** Dimensions (Student, Subject) are placed on Rows and Columns, the Marks measure controls colour and a second measure can control mark size. The mark type is Square or Show Me > Heat Map.
- **Overall average:** Total marks = 227 + 242 + 228 = 697 over 9 observations. Average = 697 / 9 = 77.44. Student averages are P = 75.67, Q = 80.67 and R = 76.00.
- **Subject averages:** Data Structures = (78 + 90 + 66) / 3 = 78.00; DBMS = (85 + 70 + 74) / 3 = 76.33; Networks = (64 + 82 + 88) / 3 = 78.00. DBMS is the weakest subject overall and the other two are equal.
- **Highest and lowest:** The highest mark is 90 (Student Q, Data Structures), shown as the darkest square. The lowest mark is 64 (Student P, Networks), shown as the lightest square. Student R performs best in Networks (88).
- **Tableau steps:** Pivot the subject columns into Subject and Marks, drag Student to Rows and Subject to Columns, select Square, drag Marks to Color with a sequential palette (light yellow to dark red) and enable labels and tooltips.

Q17. A survey of 800 students on their mode of transport gives: Bus 280, Two-wheeler 220, Walking 180, Car 120. Calculate the percentage and central angle of each category and explain how the pie chart is created in Tableau.

Introduction: A pie chart is a circular chart divided into slices, in which each slice represents the share of a category in the whole.

- **Pie chart principle:** The full circle represents 100% (360°) of the data and the size of each slice is proportional to its category value. It is used to show parts of a whole for a small number of categories.
- **Percentage:** Percentage = (Students / Total) x 100. Bus = 280 / 800 = 35%; Two-wheeler = 220 / 800 = 27.5%; Walking = 180 / 800 = 22.5%; Car = 120 / 800 = 15%. The percentages add up to 100%.
- **Central angle:** Central angle = (Students / Total) x 360°. Bus = 126°; Two-wheeler = 99°; Walking = 81°; Car = 54°. The angles add up to 360°, which verifies the calculation.
- **Tableau steps:** Drag Transport to Color, change the mark type to Pie, drag Number of Students to Angle and to Label, then apply Quick Table Calculation > Percent of Total on the label. Adjust Size and enable mark labels.
- **Interpretation and limitation:** Bus is the most popular mode (35%) and Car the least (15%). Limitation: angles are harder to compare than bar lengths and a pie with many slices is difficult to read, so a bar or donut chart may be preferred.

Q18. Explain the types of dashboard actions in Tableau with an example of each.

Introduction: Dashboard actions are interactive operations that let a user select marks in one visualisation and cause another visualisation, sheet or web page to respond.

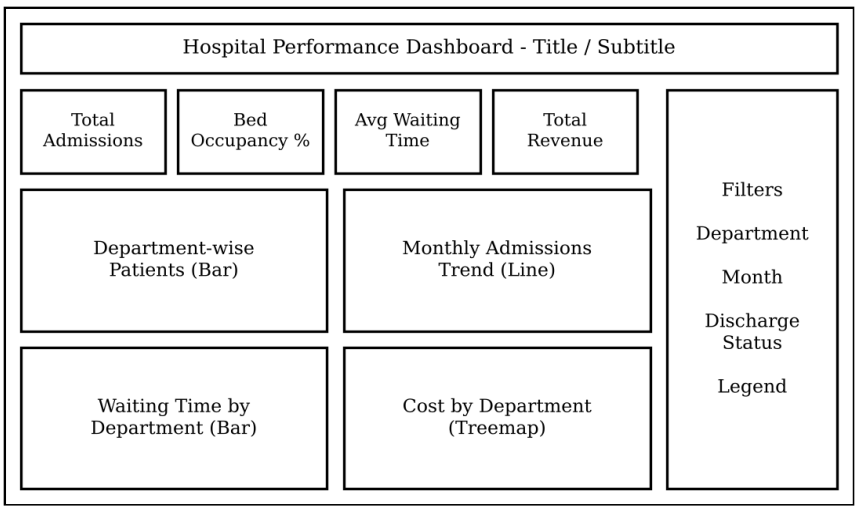
- **Filter action:** It passes the selected data from a source worksheet to filter a target worksheet. Clicking North in a region chart makes the product chart show only North sales. It is created through Dashboard > Actions > Add Action > Filter, choosing source, target and run condition.

- **Highlight action:** It emphasises the selected marks across several views while the remaining data stays visible. Selecting the Technology category highlights Technology in every worksheet, which supports comparison without removing other data.
- **URL action:** It opens an external web page when a mark is selected, using configured field values in the link. For example, clicking a product mark opens that product's information page, subject to access permissions and URL design.
- **Go to Sheet action:** It navigates from one worksheet or dashboard to another. Selecting a region in an overview dashboard opens the detailed regional dashboard. It does not filter by itself, so it is combined with a filter action for drill-through.
- **Parameter and Set actions:** Change Parameter updates a parameter value, for example selecting a chart bar to set a sales target used in calculations. Change Set Values updates the members of a set, for example adding selected customers to a customer set.

Part C - 10 Marks

Q19. A hospital wants a single Tableau dashboard to monitor its performance. The data contains Patient ID, Department, Admission Date, Doctor, Beds Occupied, Waiting Time (minutes), Treatment Cost and Discharge Status. Design the dashboard by explaining the objectives, worksheets, layout, filters, actions, formatting and testing.

Introduction: A hospital dashboard is an operational dashboard that combines KPIs and charts on one interactive screen so that management can monitor admissions, bed use, waiting time and cost together. A proposed layout is shown below.



- **Objective and audience:** The aim is to monitor admissions, bed occupancy, patient waiting time and treatment revenue for hospital management and ward staff. Defining the purpose first decides which KPIs, charts and filters are included and avoids unnecessary visuals.
- **Data preparation:** Connect Tableau to the Excel or database source and verify data types, date fields and measures. Department, Doctor and Discharge Status are dimensions, while Waiting Time, Treatment Cost and Beds Occupied are measures; remove irrelevant fields.
- **Worksheets to create:** Build separate worksheets for Total Admissions (KPI), Bed Occupancy (KPI), Department-wise Patients (bar chart), Monthly Admissions Trend (line chart), Average Waiting Time by Department (bar chart) and Cost by Department (treemap).
- **Dashboard size:** Use Dashboard > New Dashboard and select a size. Automatic adapts to the screen, Fixed (for example 1200 x 800 pixels) keeps the layout constant, and Range works between minimum and maximum limits; separate tablet and phone layouts can be added.

- **Layout and hierarchy:** Use a tiled layout: title at the top, KPI cards in the first row, trend and department charts in the middle, waiting time and cost at the bottom, and filters in a side panel. A floating object may place the hospital logo at a corner.
- **Chart selection:** Choose charts to match the question: bars for comparison across departments, a line for monthly trend, KPI cards for single values and a treemap for cost contribution. Avoid information overload by showing only the visuals needed for the objective.
- **Filters:** Add Department, Month and Discharge Status filters as multiple-value dropdowns through the worksheet menu on the dashboard, and apply them to the relevant worksheets. Keep the number of filters small because too many filters add complexity and slow performance.
- **Actions and tooltips:** Add a filter action so that selecting a department updates the trend and waiting-time charts, a highlight action for emphasis and a Go to Sheet action to a detailed patient list. Tooltips show exact values without using permanent space.
- **Formatting:** Use consistent fonts, number formats (₹ for cost, % for occupancy) and a limited colour palette, for example green for good and red for warning. Add a meaningful title with the period, legends, padding and blank objects for clean spacing.
- **Testing, performance and saving:** Check that filters, actions and layouts work and that charts are readable on each device size. Use extracts and fewer worksheets for speed, protect patient data through access control, and save as .twb or packaged .twbx workbook.

Q20. Explain the major elements of a Tableau dashboard with suitable examples.

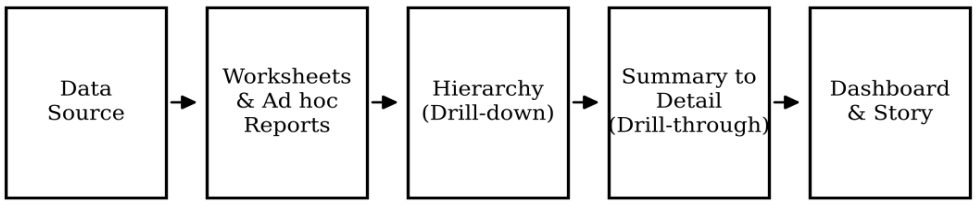
Introduction: Dashboard elements are the objects and visual components used to build, organise, format and make a Tableau dashboard interactive.

- **Worksheets:** Worksheets are the most important element; each contains a visualisation such as a bar chart, line chart, map or text table created from dimensions and measures. A hospital dashboard may include admission, department and bed-availability worksheets.
- **Text and Image objects:** Text objects add titles, subtitles, instructions and reporting periods, for example "Annual Student Performance Analysis 2025-26". Image objects insert logos or graphics, such as a university logo at the top-left corner, giving the dashboard an identity.
- **Web Page object:** It displays web content inside the dashboard, such as an organisation's website or an online report. Whether the page appears depends on browser security restrictions, authentication and the page's embedding policy.
- **Blank object:** A blank object is an empty component used only to create space between charts, separate sections and improve alignment. It avoids overcrowding without adding any information.
- **Horizontal and Vertical containers:** A horizontal container places objects side by side, such as Sales by Region and Profit by Region, while a vertical container stacks them, such as title, KPI cards and trend chart. Containers keep alignment and spacing consistent.
- **Filters:** Filters restrict the displayed data to selected values such as a region or date. Types include dimension, measure, date, context, data source and extract filters, displayed as dropdowns, lists or sliders depending on the field.
- **Legends:** A legend explains the meaning of colours, sizes or shapes, for example green for positive profit and red for negative profit. A clear legend improves chart interpretation and reduces confusion between categories.
- **Parameters:** A parameter is a dynamic value that users control, such as choosing Sales, Profit or Quantity to be displayed. It affects the dashboard only when connected to a calculated field, filter or reference line; it is not a filter by itself.
- **Actions and Navigation objects:** Actions (filter, highlight, URL, Go to Sheet, parameter and set actions) make selections in one view affect others. Navigation buttons let users move between dashboards, for example from a Sales Dashboard to a Profit Dashboard.

- **KPI cards and Tooltips:** KPI cards display key figures prominently, for example Total Revenue ₹50,00,000 or Total Customers 5,000. Tooltips are small boxes that appear on hovering over a mark and give extra details such as region, sales, profit and orders.

Q21. An e-commerce company wants a Tableau reporting system. The dataset has Order ID, Order Date, Region, Category, Sub-Category, Product, Customer, Sales and Profit. Explain how you would build a story, ad hoc reports, a drill-down hierarchy, a drill-through report and a final dashboard.

Introduction: An integrated reporting system combines a story for communicating findings, ad hoc reports for on-demand questions, drill-down and drill-through for exploring detail, and a dashboard for monitoring. The workflow is shown below.



- **Data preparation:** Connect to the Excel or CSV source and check field names, data types, date fields and measures. Identify dimensions (Region, Category, Product, Customer) and measures (Sales, Profit) so that all later worksheets are built from clean, consistent data.
- **Story design:** Define the audience and arrange story points from overview to conclusion: total sales KPI, monthly trend (line), regional sales (bar), category performance (treemap), profit analysis (scatter or bar), top products (sorted bar) and a closing recommendation view.
- **Creating the story:** Choose Story > New Story, set the size (Automatic, Fixed or Range), drag a worksheet onto the canvas, enter a caption that explains the finding and use New Blank Point for each next stage. Preview the navigation and save the workbook.
- **Ad hoc reports:** For a question such as "Which five products earned the highest sales?", place Product and Sales in the view, open the Product filter, choose Top > By Field > Top 5 by SUM(Sales) and sort. Other questions include region-wise sales and month-wise trend.
- **Calculated field for ad hoc analysis:** A derived measure such as Average Order Value is created as shown below. Expected output: if total sales are ₹1,20,000 from 8 distinct orders, the result is 1,20,000 / 8 = ₹15,000 per order.

```
SUM([Sales]) / COUNTD([Order ID])
```

- **Drill-down hierarchy:** In the Data pane right-click Category, choose Hierarchy > Create Hierarchy, name it Product Hierarchy and add Sub-Category and Product below it. The order must be Category, Sub-Category, then Product so each level is a subset of the one above.
- **Drill-down view:** Place the hierarchy on Rows and Sales on Columns with a bar chart, then use the plus sign to expand from category to sub-category to product and the minus sign to collapse. Totals of child levels must reconcile with the parent total.
- **Drill-through report:** Create a Regional Sales Summary worksheet and a Regional Transaction Details worksheet listing Order ID, Customer, Product, Sales and Profit. Add a filter action (on Select, matching Region) and a Go to Sheet action, so selecting West opens only West orders.
- **Final dashboard:** Combine KPI cards, the regional bar, monthly trend and category treemap on a tiled dashboard, with Region and Date filters, a clear title and a navigation button to the detail report. Test that every selection leads to the expected records.

- **Validation and best practice:** Verify totals against the source, check date levels and aggregations, restrict sensitive customer data to authorised users and document reports that become recurring. Drill-down alone does not explain causes, so conclusions should be stated carefully.

Q22. Explain the following Tableau chart types with their purpose and construction: Gantt, Waterfall, Donut, Funnel, Pareto, Gauge, Bump, Pyramid, Histogram and Word Cloud.

Introduction: Tableau supports specialised charts for schedules, financial changes, proportions, process stages, priorities, KPIs, rankings, demographic structures, distributions and text, each answering a different analytical question.

- **Gantt chart:** It shows the start and duration of tasks as horizontal bars on a timeline, used for project planning and scheduling. Build it with Start Date on Columns, Task on Rows, Gantt Bar as the mark and Duration on Size.
- **Waterfall chart:** It shows how an initial value reaches a final value through positive and negative contributions, such as revenue to net profit. Build it with a Running Total table calculation and Gantt bars, coloured by increase or decrease.
- **Donut chart:** It is a pie chart with a blank centre that can show a total or KPI, used for category contribution. Tableau builds it by a dual axis: a pie on one axis and a smaller background-coloured circle (using MIN(1)) overlaid on the other.
- **Funnel chart:** It shows records remaining at sequential stages and reveals drop-off, for example visitors to purchases. Place Stage on Rows and the count on Columns, sort by process order and compute the conversion rate between stages.
- **Pareto chart:** It combines bars sorted in descending order with a cumulative-percentage line to identify the few categories giving most of the total (the 80/20 guideline). Apply Running Total and Percent of Total calculations on a dual axis.
- **Gauge chart:** It displays a KPI against a target or range using a semicircular pointer, such as 80% target achievement. It needs custom construction with calculated fields, pie marks and a dual axis; a bullet chart is often a simpler alternative.
- **Bump chart:** It shows how the rank of entities changes over time, with one line per entity and rank on the vertical axis. Use RANK(SUM([Sales]), 'desc') as a table calculation computed per period and reverse the axis so rank 1 is on top.
- **Pyramid chart:** It compares two mirrored groups across ordered categories, as in a population pyramid by age. Convert one group to negative values with an IF calculation, place Age Group on Rows and the field on Columns with Group on Color.
- **Histogram:** It shows the frequency distribution of a continuous variable such as student marks across bins. Create Bins on the measure, place the bin on Columns and Number of Records on Rows; the shape shows peaks, skewness and gaps.
- **Word cloud:** It displays words sized by frequency, for example customer feedback terms. Prepare a Word and Frequency table after removing common words, drag Word to Text and Frequency to Size; frequency alone does not show sentiment.

Q23. The delivery times (in minutes) of 12 orders are: 8, 11, 13, 15, 17, 19, 21, 24, 26, 29, 33, 68. Calculate Q1, median, Q3, IQR, the outlier limits and the whiskers, identify any outliers, and explain how the box plot is created in Tableau.

Introduction: A box plot (box-and-whisker plot) summarises the distribution of numerical data using the median, quartiles, whiskers and outliers, and is used to compare spread and unusual values across categories.

- **Components:** The box extends from Q1 (25th percentile) to Q3 (75th percentile) with the median as a line inside; the IQR = Q3 - Q1 is the spread of the middle 50%. Whiskers reach the smallest and largest non-outlier values, and points beyond them are outliers.

- **Median (Q2):** The data are already sorted and n = 12 (even), so the median is the average of the 6th and 7th values: Q2 = (19 + 21) / 2 = 20 minutes. This is the typical delivery time and splits the data into two halves.
- **First quartile (Q1):** The lower half is 8, 11, 13, 15, 17, 19. Its median is the average of the 3rd and 4th values, so Q1 = (13 + 15) / 2 = 14 minutes, using the median-of-halves method.
- **Third quartile (Q3):** The upper half is 21, 24, 26, 29, 33, 68. Its median is the average of the 3rd and 4th values, so Q3 = (26 + 29) / 2 = 27.5 minutes.
- **Interquartile range:** IQR = Q3 - Q1 = 27.5 - 14 = 13.5 minutes. The middle 50% of orders are delivered within a spread of 13.5 minutes, and this value is used to fix the outlier limits.
- **Outlier limits:** Lower limit = Q1 - 1.5(IQR) = 14 - 20.25 = -6.25; upper limit = Q3 + 1.5(IQR) = 27.5 + 20.25 = 47.75. Any value below -6.25 or above 47.75 is flagged as a potential outlier.
- **Outliers and whiskers:** The value 68 exceeds 47.75, so it is the only outlier. The lower whisker ends at 8 and the upper whisker at 33, the smallest and largest non-outlier values; the whiskers do not reach the absolute maximum of 68.
- **Tableau steps:** Connect to the data, drag a record-level field such as Order ID to Detail so individual values exist, place Delivery Time on Rows, choose Show Me > Box-and-Whisker Plot, or drag Box Plot from the Analytics pane, and format axes and tooltips.
- **Interpretation:** The box spans 14 to 27.5 with the median at 20, near the centre, so the middle half of the orders is fairly balanced. The single order of 68 minutes is exceptionally slow and should be investigated as a possible delay.
- **Precautions and use:** Quartile values can differ slightly between software conventions, and a view with one aggregated mark per category shows no meaningful distribution. Box plots are widely used to compare marks, salaries and response times across groups.

Q24. A company has the following quarterly sales data. (1) Calculate the total annual sales of each product. (2) Find the quarter with the highest total sales. (3) Calculate the percentage increase in Product R sales from Q1 to Q4. (4) Identify the most suitable visualisation (bar chart, line chart or pie chart) for the quarterly trend and justify it. (5) Explain how the chosen chart is created in Tableau.

Quarter	Product P (₹)	Product Q (₹)	Product R (₹)
Q1	30,000	22,000	18,000
Q2	34,000	26,000	21,000
Q3	29,000	31,000	27,000
Q4	42,000	36,000	33,000

Introduction: This is an analysis problem using aggregation, percentage change and chart selection, where the totals are first calculated from the data and then interpreted to choose an appropriate visualisation.

- **(1) Annual totals:** Product P = 30,000 + 34,000 + 29,000 + 42,000 = ₹1,35,000; Product Q = 22,000 + 26,000 + 31,000 + 36,000 = ₹1,15,000; Product R = 18,000 + 21,000 + 27,000 + 33,000 = ₹99,000. The grand total is ₹3,49,000.
- **Share of products:** Product P contributes 1,35,000 / 3,49,000 = 38.7%, Product Q 33.0% and Product R 28.4% of annual sales. Product P is the largest contributor and Product R the smallest, though R shows the fastest growth.
- **(2) Quarter totals:** Q1 = 30,000 + 22,000 + 18,000 = ₹70,000; Q2 = 34,000 + 26,000 + 21,000 = ₹81,000; Q3 = 29,000 + 31,000 + 27,000 = ₹87,000; Q4 = 42,000 + 36,000 + 33,000 = ₹1,11,000. Totals add up to ₹3,49,000.
- **Highest quarter:** Q4 has the highest total sales of ₹1,11,000, about 31.8% of the annual total, and totals increase in every successive quarter. Q4 is therefore the strongest quarter for all three products combined.

- **(3) Percentage increase:** Percentage increase = [(Q4 - Q1) / Q1] x 100. For Product R = [(33,000 - 18,000) / 18,000] x 100 = (15,000 / 18,000) x 100 = 83.33%.
- **Comparison:** For comparison, Product Q increased by (36,000 - 22,000) / 22,000 = 63.64% and Product P by (42,000 - 30,000) / 30,000 = 40%. Product R is therefore the fastest-growing product between Q1 and Q4.
- **(4) Suitable chart:** A line chart is the most suitable because the requirement is a quarterly trend over time. Quarters are placed in sequence on the X axis and one line per product shows direction, slope and any dip, such as Product P falling in Q3.
- **Why not bar or pie:** A pie chart shows composition at a single point in time and cannot show change across four quarters. A bar chart can compare values but does not show continuous direction of change as clearly, while a stacked bar suits composition rather than trend.
- **(5) Creation in Tableau:** Pivot the product columns into Product and Sales fields, drag Quarter to Columns and Sales to Rows, select Line on the Marks card and drag Product to Color. Add Sales to Label, sort quarters Q1 to Q4 and format values in rupees.
- **Interpretation:** All products rise strongly in Q4, Product P dips in Q3 and Product Q grows steadily. The chart shows what happened but not why, so conclusions must not claim causes unsupported by the data; a dashboard or story can present these findings.